

Rafi Rafsan

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 Gazipur, Bangladesh

Profile

Computer Science graduate with research and development experience in data analytics, applied machine learning, and systems analysis. Skilled in designing data processing pipelines, building AI-driven advisory tools, and evaluating user acceptance. Experienced in field-level volunteering, quantitative/qualitative data practices, and analyzing development-sector operational workflows.

Research Experience & Technical Writing

● Undergraduate Thesis

Hyperspectral Image (HSI) Classification

- A transformer-based feature extraction model is used to classify HSI pixels.
- Using only 5 labeled samples, achieved 86.78% accuracy on cross domain dataset.
- Analyzed the influence of source domain selection in cross-domain classification.

● SemEval-2026 Shared Task

Political question-evasion classification

- Used a generative classification pipeline based on different prompting strategy (few shot, COT) .
- Experimented with a Natural Language Inference (NLI) framework utilizing Longformer-base for data analysis.
- Addressed the class imbalances issue using a weighted cross-entropy loss function, improving the minority class F1-score from 0.53 to 0.71.

● Technical Writing Collaborative Research Study

User Experience and Acceptance of AI

- Conducted a literature-based analysis of user acceptance, friction points, and cognitive load factors across AI systems.
- Demonstrated Quantitative and qualitative findings regarding usability, data privacy, and trust factors.
- Compiled actionable design principles.

Education

● Computer Science and Engineering

Rajshahi University of Engineering & Technology

Graduation: May 2025

CGPA: 3.51/4.00

Projects

● Porichorja – AI-Powered Agricultural Advisory System

Tools Used: Python, FastAPI, Ollama, Vector Database, Embedding Models, RAG Pipeline

- Collected, compiled, and structured a domain-specific dataset of local agricultural queries.
- Designed a data preprocessing pipeline featuring text normalization, cleaning, and text-document chunking.
- Integrated an embedding-based semantic search system with a vector database to deploy a Retrieval-Augmented Generation (RAG) tool
- Configured local Large Language Model (LLM) inference via the Ollama framework for contextual query answering

● Information System Analysis and Design Report on BRAC

Focus Areas: Systems Analysis, Field Frameworks, Workflow Mapping

- Analyzed operational workflows and monitoring systems within BRAC's Urban Development Program (UDP).
- Documented field-level stakeholder engagement, data management practices, and requirements gathering protocols.
- Evaluated the use of ICT-based data collection methods and field data workflows alongside existing monitoring and evaluation (M&E) frameworks

● Sarangsho – Book Review Platform

Technologies: FastAPI, JWT, SQLAlchemy, Python

- Designed RESTful APIs with full CRUD functionality and relational review system.
- Implemented JWT-based authentication and role-based access control.
- Developed advanced search, filtering, and dynamic rating computation.

Analytical Expertise

- Solved 500+ algorithmic problems across competitive programming platforms.
 - Codeforces: musafeer
 - LeetCode: rafi 1903139

Volunteering

● Shojon Foundation IT Support

- Managed field-level distribution logistics for emergency flash-flood relief materials in Sunamganj.
- Distributed winter clothing networks in Thakurgaon and basic necessities during COVID-19 lock-downs in Mymensingh.
- Coordinated local distribution events for primary educational supplies to children in Gazipur.
- Maintained digital tracking records of inventory, donor resources, and community allocation sheets.

Skills

● LANGUAGES

Bangla
Native

English
Intermediate

Arabic
Basic

● TECHNICAL SKILLS

Programming Languages
Python

Data Analysis
Pandas, Numpy

Data Visualization
Matplotlib, Seaborn

Machine Learning
Pytorch, OpenCV

Database Management
SQL, Postgresql

Web Framwork
FastAPI

● PROFESSIONAL SKILLS

- Data collection and documentation
- Data Analysis and visualization
- Team collaboration
- Literature review and Academic Writing

Certifications:

- **Technical Support Fundamentals**
Google
- **Machine Learning with Python**
IBM

References

- **Fuad Al Abir**
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